

Practice Exam #1

Full-Length Timed Simulation

110 questions • 5 hours 20 minutes

Instructions: NCEES FE Reference Handbook and approved calculator only. Pace about 2.9 minutes per question. Write units at every step; never leave a blank.

Questions

1. Water ($\rho = 1000 \text{ kg/m}^3$) fills a tank to a depth of 6 m. The gauge pressure at the bottom is closest to:
(A) 29.4 kPa (B) 58.9 kPa
(C) 5.89 kPa (D) 117.7 kPa
2. A pipe carries $Q = 0.30 \text{ m}^3/\text{s}$ and has an inside diameter $D = 0.40 \text{ m}$. The average velocity is closest to:
(A) 2.39 m/s (B) 0.75 m/s
(C) 1.19 m/s (D) 4.77 m/s
3. Water discharges from a small orifice 5.0 m below the free surface of a large tank. Neglecting losses, the jet velocity is closest to:
(A) 7.0 m/s (B) 22.1 m/s
(C) 9.9 m/s (D) 4.95 m/s
4. Water ($\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$) flows at $V = 2.0 \text{ m/s}$ in a 0.10 m diameter pipe. The Reynolds number is closest to:
(A) 2,000 (B) 20,000
(C) 2×10^6 (D) 200,000
5. For a pipe with friction factor $f = 0.02$, length $L = 100 \text{ m}$, diameter $D = 0.20 \text{ m}$, carrying water at $V = 3.0 \text{ m/s}$, the Darcy–Weisbach head loss is closest to:
(A) 4.59 m (B) 9.17 m
(C) 2.29 m (D) 0.46 m
6. An open channel has hydraulic radius $R_h = 0.50 \text{ m}$, slope $S = 0.001$, and Manning $n = 0.013$. Using SI Manning’s equation, the velocity is closest to:
(A) 0.77 m/s (B) 1.53 m/s
(C) 3.06 m/s (D) 0.48 m/s
7. A rectangular channel 2.0 m wide flows 1.0 m deep ($R_h = 0.50 \text{ m}$). With $n = 0.015$ and $S = 0.0009$, the discharge Q is closest to:
(A) $0.84 \text{ m}^3/\text{s}$ (B) $2.52 \text{ m}^3/\text{s}$
(C) $1.26 \text{ m}^3/\text{s}$ (D) $3.36 \text{ m}^3/\text{s}$

8. A pump delivers $Q = 0.05 \text{ m}^3/\text{s}$ of water against a total head $H = 20 \text{ m}$ at an efficiency of 75%. The required input power is closest to:
- (A) 9.81 kW (B) 19.6 kW
(C) 7.36 kW (D) 13.1 kW
9. Two identical pumps, each producing a head of 15 m at the design flow, are connected in series. The combined head at that flow is closest to:
- (A) 30 m (B) 15 m
(C) 7.5 m (D) 60 m
10. A suppressed rectangular weir has crest length $L = 2.0 \text{ m}$ and head $H = 0.30 \text{ m}$. Using $Q = 1.84 L H^{3/2}$ (SI), the discharge is closest to:
- (A) $0.30 \text{ m}^3/\text{s}$ (B) $0.605 \text{ m}^3/\text{s}$
(C) $1.10 \text{ m}^3/\text{s}$ (D) $0.20 \text{ m}^3/\text{s}$
11. An orifice of area $A = 0.01 \text{ m}^2$ ($C_d = 0.61$) discharges under a head of 3.0 m. The discharge is closest to:
- (A) $0.122 \text{ m}^3/\text{s}$ (B) $0.077 \text{ m}^3/\text{s}$
(C) $0.0468 \text{ m}^3/\text{s}$ (D) $0.234 \text{ m}^3/\text{s}$
12. A fitting with minor-loss coefficient $K = 1.5$ passes water at $V = 4.0 \text{ m/s}$. The minor head loss is closest to:
- (A) 1.22 m (B) 0.82 m
(C) 2.45 m (D) 0.61 m
13. A flat gate has area $A = 0.20 \text{ m}^2$ and a uniform gauge pressure of 50 kPa acts on it. The resultant force is closest to:
- (A) 5.0 kN (B) 10.0 kN
(C) 2.5 kN (D) 25 kN
14. A treatment unit reduces BOD from 200 mg/L to 20 mg/L. The removal efficiency is:
- (A) 80% (B) 10%
(C) 90% (D) 95%
15. A tank of volume $V = 2000 \text{ m}^3$ receives a flow $Q = 500 \text{ m}^3/\text{h}$. The hydraulic detention time is:
- (A) 4 h (B) 0.25 h
(C) 8 h (D) 2 h
16. A clarifier treats $Q = 4000 \text{ m}^3/\text{d}$ over a surface area $A = 200 \text{ m}^2$. The surface overflow rate is:
- (A) 10 m/d (B) 20 m/d
(C) 40 m/d (D) 0.05 m/d

17. A water contains 40 mg/L Ca^{2+} and 12 mg/L Mg^{2+} . The total hardness expressed as CaCO_3 is closest to: (EW Ca = 20, EW Mg = 12.15)
- (A) 52 mg/L (B) 100 mg/L
(C) 210 mg/L (D) 149 mg/L
18. A chlorine residual of 2.0 mg/L is maintained for a contact time of 30 min. The CT value is:
- (A) 60 mg·min/L (B) 15 mg·min/L
(C) 30 mg·min/L (D) 120 mg·min/L
19. An activated-sludge plant treats $Q = 10,000 \text{ m}^3/\text{d}$ with influent BOD $S_0 = 200 \text{ mg/L}$. The aeration basin volume is 2500 m^3 at MLSS $X = 2000 \text{ mg/L}$. The F/M ratio (d^{-1}) is closest to:
- (A) 0.20 (B) 0.40
(C) 0.80 (D) 0.10
20. In the plant above the reactor holds 5000 kg of MLSS solids and 500 kg/d of solids are wasted. The solids retention time (SRT) is closest to:
- (A) 5 d (B) 2 d
(C) 10 d (D) 20 d
21. A wastewater with ultimate BOD $L_0 = 300 \text{ mg/L}$ decays at $k = 0.23 \text{ d}^{-1}$ (base e). The BOD exerted at 5 days is closest to:
- (A) 95 mg/L (B) 150 mg/L
(C) 300 mg/L (D) 205 mg/L
22. Two streams blend: $Q_1 = 2.0 \text{ m}^3/\text{s}$ at $C_1 = 10 \text{ mg/L}$ and $Q_2 = 3.0 \text{ m}^3/\text{s}$ at $C_2 = 20 \text{ mg/L}$. The mixed concentration is:
- (A) 16 mg/L (B) 15 mg/L
(C) 30 mg/L (D) 12 mg/L
23. A plant treating 5.0 MGD applies a chlorine dose of 2.0 mg/L. The chlorine feed rate is closest to:
- (A) 41.7 lb/d (B) 83.4 lb/d
(C) 167 lb/d (D) 10 lb/d
24. A water has 150 mg/L of carbonate hardness as CaCO_3 . Expressed in equivalents, this is:
- (A) 1.5 meq/L (B) 6 meq/L
(C) 3 meq/L (D) 0.3 meq/L
25. A sludge stream contains 1000 kg/d of dry solids at 5% solids by mass. Assuming the wet sludge density is $\approx 1000 \text{ kg/m}^3$, the daily wet sludge volume is closest to:
- (A) 20 m^3/d (B) 50 m^3/d
(C) 5 m^3/d (D) 200 m^3/d

26. A flow of 0.50 MGD carries a BOD of 250 mg/L. The BOD mass loading is closest to:
- (A) 521 lb/d (B) 2085 lb/d
(C) 125 lb/d (D) 1043 lb/d
27. Using the rational method (English units) with $C = 0.50$, $i = 2.0$ in/hr, and $A = 10$ acres, the peak runoff is closest to:
- (A) 5 cfs (B) 20 cfs
(C) 10 cfs (D) 1 cfs
28. Using the SI rational method $Q = CiA/360$ with $C = 0.40$, $i = 90$ mm/hr, and $A = 5$ ha, the peak runoff is closest to:
- (A) $0.50 \text{ m}^3/\text{s}$ (B) $5.0 \text{ m}^3/\text{s}$
(C) $0.05 \text{ m}^3/\text{s}$ (D) $1.8 \text{ m}^3/\text{s}$
29. Overland flow travels 300 m at an average velocity of 1.5 m/s. The travel (concentration) time for that segment is closest to:
- (A) 1.67 min (B) 3.33 min
(C) 5.0 min (D) 200 min
30. A detention basin receives $Q_{in} = 5.0 \text{ m}^3/\text{s}$ and releases $Q_{out} = 2.0 \text{ m}^3/\text{s}$ for 1.0 hour. The required storage volume is closest to:
- (A) 3600 m^3 (B) 7200 m^3
(C) $18,000 \text{ m}^3$ (D) $10,800 \text{ m}^3$
31. A stream has a dissolved-oxygen saturation of 9.0 mg/L and an actual DO of 6.0 mg/L. The initial oxygen deficit is:
- (A) 3.0 mg/L (B) 6.0 mg/L
(C) 9.0 mg/L (D) 1.5 mg/L
32. Using Streeter-Phelps with $k_d = 0.30 \text{ d}^{-1}$, $k_r = 0.40 \text{ d}^{-1}$, $L_0 = 20 \text{ mg/L}$, and initial deficit $D_0 = 2.0 \text{ mg/L}$, the DO deficit at $t = 2.0 \text{ d}$ is closest to:
- (A) 2.0 mg/L (B) 4.5 mg/L
(C) 6.87 mg/L (D) 9.0 mg/L
33. For the stream of the previous problem, the critical (maximum-deficit) time t_c is closest to:
- (A) 1.0 d (B) 2.54 d
(C) 5.0 d (D) 0.25 d
34. For a watershed, annual precipitation is 1000 mm, evapotranspiration 600 mm, and runoff 300 mm. The change in storage is:
- (A) 100 mm (B) 200 mm
(C) 400 mm (D) 700 mm

35. A design storm has an intensity of 50 mm/hr lasting 0.50 hr. The total rainfall depth is:
- (A) 100 mm (B) 50 mm
(C) 25 mm (D) 12.5 mm
36. A 10,000 m² catchment with runoff coefficient $C = 0.60$ receives 0.050 m of rain. The runoff volume is closest to:
- (A) 100 m³ (B) 500 m³
(C) 150 m³ (D) 300 m³
37. An aquifer has hydraulic conductivity $K = 1.0 \times 10^{-4}$ m/s and gradient $i = 0.02$. The Darcy (specific) discharge is:
- (A) 5×10^{-6} m/s (B) 2×10^{-6} m/s
(C) 2×10^{-4} m/s (D) 5×10^{-3} m/s
38. For the aquifer above with porosity $n = 0.25$, the average linear (seepage) velocity is:
- (A) 8×10^{-6} m/s (B) 2×10^{-6} m/s
(C) 5×10^{-7} m/s (D) 3.2×10^{-5} m/s
39. Two wells 500 m apart show a head difference of 2.0 m. The hydraulic gradient is:
- (A) 0.04 (B) 250
(C) 0.004 (D) 0.4
40. A confined aquifer has $K = 15$ m/d and thickness $b = 20$ m. The transmissivity is:
- (A) 35 m²/d (B) 0.75 m²/d
(C) 150 m²/d (D) 300 m²/d
41. Groundwater flows through a cross-sectional area $A = 50$ m² with $K = 1.0 \times 10^{-3}$ m/s and $i = 0.01$. The volumetric flow is:
- (A) 5×10^{-4} m³/s (B) 5×10^{-3} m³/s
(C) 5×10^{-5} m³/s (D) 1×10^{-3} m³/s
42. For confined steady radial flow (Thiem), $Q = 2\pi T(h_2 - h_1)/\ln(r_2/r_1)$. With $T = 300$ m²/d, $h_2 - h_1 = 3.0$ m, and $r_2/r_1 = 10$, Q is closest to:
- (A) 1230 m³/d (B) 820 m³/d
(C) 2456 m³/d (D) 410 m³/d
43. A contaminant has distribution coefficient $K_d = 2.0$ cm³/g in soil with bulk density $\rho_b = 1.6$ g/cm³ and porosity $n = 0.30$. The retardation factor $R = 1 + \rho_b K_d/n$ is closest to:
- (A) 5.7 (B) 11.67
(C) 1.0 (D) 3.2

44. A Darcy flux of 2×10^{-6} m/s acts over an aquifer thickness of 20 m. The groundwater flow per unit width of aquifer is:
- (A) 4×10^{-5} m²/s (B) 2×10^{-6} m²/s
(C) 1×10^{-4} m²/s (D) 4×10^{-7} m²/s
45. A nonsorbing tracer travels 80 m at a seepage velocity of 8×10^{-6} m/s. The travel time is closest to:
- (A) 10 d (B) 50 d
(C) 1000 d (D) 115.7 d
46. At 25 °C and 1 atm (molar volume 24.45 L/mol), 0.10 ppm of SO₂ (MW 64) corresponds to a mass concentration of:
- (A) 64 µg/m³ (B) 131 µg/m³
(C) 262 µg/m³ (D) 655 µg/m³
47. At 25 °C and 1 atm, 5725 µg/m³ of CO (MW 28) corresponds to:
- (A) 2.5 ppm (B) 5.0 ppm
(C) 10 ppm (D) 1.0 ppm
48. A process emits a pollutant with an emission factor of 10 kg per Mg of product and produces 500 Mg/d. The emission rate is:
- (A) 5000 kg/d (B) 50 kg/d
(C) 500 kg/d (D) 50,000 kg/d
49. Under dry adiabatic conditions (lapse rate 9.8 °C/km), a parcel rising 2.0 km cools by about:
- (A) 4.9 °C (B) 9.8 °C
(C) 29.4 °C (D) 19.6 °C
50. An electrostatic precipitator has drift velocity $w = 0.10$ m/s, collection area $A = 200$ m², and gas flow $Q = 10$ m³/s. By the Deutsch equation $\eta = 1 - e^{-wA/Q}$, the efficiency is closest to:
- (A) 0.50 (B) 0.63
(C) 0.865 (D) 0.95
51. For the same ESP gas flow ($Q = 10$ m³/s) and drift velocity ($w = 0.10$ m/s), the collection area required for 99% efficiency is closest to:
- (A) 461 m² (B) 200 m²
(C) 100 m² (D) 921 m²
52. A baghouse filters $Q = 600$ m³/min through $A = 300$ m² of fabric. The air-to-cloth ratio is:
- (A) 0.5 m/min (B) 2.0 m/min
(C) 1.0 m/min (D) 5.0 m/min

53. A stack emits a pollutant at 2.0 g/s into an air stream of 100 m³/s. The fully mixed concentration is:
- (A) 2 mg/m³ (B) 5 mg/m³
(C) 20 mg/m³ (D) 200 mg/m³
54. A ground-level source ($H = 0$) emits $Q = 100$ g/s. At a point where $\sigma_y = 30$ m, $\sigma_z = 15$ m and wind speed $u = 4.0$ m/s, the centerline concentration $C = Q/(\pi\sigma_y\sigma_z u)$ is closest to:
- (A) 0.0177 g/m³ (B) 0.0354 g/m³
(C) 0.0088 g/m³ (D) 0.0708 g/m³
55. A solution contains 100 mg/L of CaCO₃ (MW 100). Its molar concentration is:
- (A) 1×10^{-2} M (B) 1×10^{-3} M
(C) 1×10^{-4} M (D) 1.0 M
56. A solution has $[H^+] = 1 \times 10^{-4}$ mol/L. The pH is:
- (A) 10 (B) 7
(C) 4 (D) 1×10^{-4}
57. A solution has pH = 9.0. Its pOH is:
- (A) 5.0 (B) 9.0
(C) 14.0 (D) 4.5
58. For the pH = 9.0 solution above, the hydroxide concentration $[OH^-]$ is:
- (A) 1×10^{-9} M (B) 1×10^{-7} M
(C) 1×10^{-14} M (D) 1×10^{-5} M
59. A compound degrades by first-order kinetics with $k = 0.05$ d⁻¹. Its half-life is closest to:
- (A) 6.93 d (B) 13.86 d
(C) 20 d (D) 3.47 d
60. A first-order reaction starts at $C_0 = 50$ mg/L with $k = 0.10$ d⁻¹. After 10 days the concentration is closest to:
- (A) 25 mg/L (B) 5 mg/L
(C) 18.4 mg/L (D) 45 mg/L
61. A volatile compound has a dimensionless Henry's constant $H_c = C_g/C_l = 0.10$. If the aqueous concentration is 2.0 mg/L, the equilibrium gas-phase concentration is:
- (A) 0.2 mg/m³·(per mg/L) (B) 20 (per mg/L)
(C) 2.0 (per mg/L) (D) 0.02 (per mg/L)

62. For complete combustion $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, the mass of O_2 required per mole of CH_4 is:
- (A) 16 g (B) 32 g
(C) 48 g (D) 64 g
63. A town of 10,000 people grows arithmetically by 200 people per year. The projected population in 10 years is:
- (A) 10,200 (B) 12,000
(C) 20,000 (D) 11,000
64. A town of 10,000 grows geometrically at 3.0%/yr. The projected population in 10 years is closest to:
- (A) 13,000 (B) 10,300
(C) 13,439 (D) 30,000
65. A city uses 4.0 MGD for 20,000 people. The per-capita demand is:
- (A) 200 gpcd (B) 100 gpcd
(C) 500 gpcd (D) 50 gpcd
66. A first-order reaction ($k = 0.5 \text{ h}^{-1}$) runs in a plug-flow reactor with residence time $\tau = 2.0 \text{ h}$. The fraction remaining C/C_0 is closest to:
- (A) 0.50 (B) 0.135
(C) 0.607 (D) 0.368
67. The same reaction ($k = 0.5 \text{ h}^{-1}$, $\tau = 2.0 \text{ h}$) in a single CSTR gives $C/C_0 = 1/(1 + k\tau)$, which is:
- (A) 0.50 (B) 0.368
(C) 0.33 (D) 0.25
68. A rod carries an axial load of 10,000 N over a cross-sectional area of 0.0020 m^2 . The normal stress is:
- (A) 2.5 MPa (B) 5.0 MPa
(C) 5.0 kPa (D) 20 MPa
69. A steel member ($E = 200 \text{ GPa}$) is stressed to 200 MPa within the elastic range. The axial strain is:
- (A) 1×10^{-2} (B) 1×10^{-4}
(C) 1×10^{-3} (D) 1.0
70. A community of 5000 people has a per-capita demand of 200 gpcd. The design average flow is:
- (A) 0.2 MGD (B) 2.0 MGD
(C) 5.0 MGD (D) 1.0 MGD

71. A community of 10,000 generates MSW at 5.0 lb/person/day. The daily generation is:
- (A) 25 tons/d (B) 50 tons/d
(C) 2.5 tons/d (D) 250 tons/d
72. Loose MSW at 100 kg/m³ is compacted to 500 kg/m³. The ratio of compacted to loose volume is:
- (A) 5.0 (B) 0.5
(C) 0.2 (D) 0.8
73. A volume of 1000 m³ of loose waste (100 kg/m³) is compacted to 500 kg/m³. The compacted volume is:
- (A) 100 m³ (B) 200 m³
(C) 500 m³ (D) 5000 m³
74. A landfill produces 200 m³ of gas that is roughly 50% CH₄ / 50% CO₂ by volume. The methane volume is about:
- (A) 50 m³ (B) 150 m³
(C) 200 m³ (D) 100 m³
75. Over a 1000 m² landfill cell, infiltration is precipitation 1.0 m minus evapotranspiration 0.40 m minus runoff 0.20 m. The leachate (infiltration) volume is:
- (A) 400 m³ (B) 600 m³
(C) 1000 m³ (D) 200 m³
76. A program diverts 8 tons of a 40-ton waste stream for recycling. The diversion rate is:
- (A) 40% (B) 8%
(C) 20% (D) 80%
77. 10 tons of MSW has a heating value of 5000 BTU/lb. The total energy content is:
- (A) 5×10^7 BTU (B) 1×10^8 BTU
(C) 1×10^7 BTU (D) 5×10^8 BTU
78. A landfill receives 100 tons/d of waste compacted to an in-place density of 0.50 ton/m³. The daily airspace consumed is:
- (A) 50 m³/d (B) 500 m³/d
(C) 100 m³/d (D) 200 m³/d
79. The larger root of $x^2 - 5x + 6 = 0$ is:
- (A) 2 (B) 3
(C) 6 (D) 1

80. The value of $\sin 30^\circ + \cos 60^\circ$ is:
(A) 1.0 (B) 0.5
(C) 1.5 (D) 0.0
81. The derivative of $f(x) = 3x^2$ evaluated at $x = 2$ is:
(A) 6 (B) 4
(C) 12 (D) 24
82. The definite integral $\int_0^2 3x^2 dx$ equals:
(A) 4 (B) 6
(C) 12 (D) 8
83. The distance between the points $(0, 0)$ and $(3, 4)$ is:
(A) 5 (B) 7
(C) 25 (D) 3.5
84. The solution of $2^x = 16$ is:
(A) 2 (B) 4
(C) 8 (D) 3
85. Under the NCEES Model Rules, the engineer's foremost obligation is to:
(A) hold paramount the safety, health, and welfare of the public (B) maximize the client's profit
(C) complete the project on schedule (D) follow the employer's instructions without question
86. An engineer should perform services only in areas of their:
(A) highest billing rate (B) competence
(C) personal interest (D) employer's choosing
87. Which federal law primarily regulates discharges of pollutants into navigable surface waters of the United States?
(A) Safe Drinking Water Act (SDWA) (B) Clean Air Act (CAA)
(C) Clean Water Act (CWA) (D) Resource Conservation and Recovery Act (RCRA)
88. CERCLA (Superfund) is best described as the law governing:
(A) cleanup of abandoned/uncontrolled hazardous waste sites and liability for releases (B) cradle-to-grave management of active hazardous waste
(C) drinking-water maximum contaminant levels (D) ambient air-quality standards

89. An engineer discovers that a design flaw endangers the public but the client refuses to address it. The most ethical action is to:
- (A) ignore it since the client decides (B) redesign without telling anyone
(C) resign quietly (D) notify the proper authorities and the client as required to protect the public
90. Which statute requires federal agencies to prepare an Environmental Impact Statement (EIS) for major actions significantly affecting the environment?
- (A) RCRA (B) NEPA
(C) OSHA (D) SDWA
91. A future amount of \$10,000 is due in 10 years at an interest rate of 5%. Its present worth (P/F) is closest to:
- (A) \$5000 (B) \$9500
(C) \$6139 (D) \$16,289
92. The present worth of an annuity of \$1000/yr for 5 years at 8% (P/A) is closest to:
- (A) \$3993 (B) \$5000
(C) \$4500 (D) \$3000
93. An investment of \$20,000 is recovered uniformly over 10 years at 6% (A/P). The annual payment is closest to:
- (A) \$2000 (B) \$2717
(C) \$3200 (D) \$1200
94. Equipment costs \$50,000, has a salvage value of \$5000, and a 9-year life. The straight-line annual depreciation is:
- (A) \$5556 (B) \$4500
(C) \$5500 (D) \$5000
95. A project has present-worth benefits of \$150,000 and present-worth costs of \$100,000. The benefit-cost ratio is:
- (A) 1.5 (B) 0.67
(C) 2.5 (D) 1.0
96. A product sells for \$5/unit with variable cost \$3/unit and fixed cost \$10,000. The break-even quantity is:
- (A) 2000 units (B) 3333 units
(C) 5000 units (D) 10,000 units
97. For the data set $\{2, 4, 4, 4, 5, 5, 7, 9\}$, the mean is:
- (A) 4 (B) 5
(C) 6 (D) 4.5

98. For the data set $\{2, 4, 4, 4, 5, 5, 7, 9\}$ (mean 5), the population standard deviation is:
- (A) 2.0 (B) 2.14
(C) 4.0 (D) 1.41
99. A normally distributed variable has $\mu = 50$ and $\sigma = 10$. The z -score of an observation $x = 70$ is:
- (A) 0.5 (B) 1.0
(C) 7.0 (D) 2.0
100. For the data set $\{2, 4, 4, 4, 5, 5, 7, 9\}$, the sample ($n-1$) standard deviation is closest to:
- (A) 2.0 (B) 4.57
(C) 2.14 (D) 1.51
101. Drinking water contains 0.050 mg/L of a contaminant. For an adult ingesting 2.0 L/d with body weight 70 kg, the chronic daily intake $CDI = C \cdot IR/BW$ is closest to:
- (A) 0.0007 mg/kg-d (B) 0.00143 mg/kg-d
(C) 0.0143 mg/kg-d (D) 0.10 mg/kg-d
102. For the same exposure ($CDI = 0.00143$ mg/kg-d) and a cancer slope factor $SF = 0.50$ per mg/kg-d, the incremental lifetime cancer risk is closest to:
- (A) 7.1×10^{-4} (B) 1.4×10^{-3}
(C) 7.1×10^{-5} (D) 3.6×10^{-4}
103. A noncarcinogen has an intake of 0.0020 mg/kg-d and a reference dose $RfD = 0.010$ mg/kg-d. The hazard quotient is:
- (A) 5.0 (B) 2.0
(C) 0.20 (D) 0.020
104. An adult ingests 2.0 L/d of water containing 10 mg/L of a substance. The daily intake mass is:
- (A) 5 mg/d (B) 10 mg/d
(C) 50 mg/d (D) 20 mg/d
105. A household uses 1000 kWh of electricity with a grid emission factor of 0.50 kg CO₂/kWh. The carbon footprint is:
- (A) 2000 kg CO₂ (B) 500 kg CO₂
(C) 50 kg CO₂ (D) 1500 kg CO₂
106. Using a 100-year global warming potential of 25 for methane, 2.0 kg of CH₄ is equivalent to:
- (A) 2 kg CO₂e (B) 25 kg CO₂e
(C) 50 kg CO₂e (D) 27 kg CO₂e

107. Burning 100 L of diesel (energy content 36 MJ/L) releases an energy of:
- (A) 3600 MJ (B) 360 MJ
(C) 36,000 MJ (D) 136 MJ
108. Using $PV = nRT$ with $n = 2.0$ mol, $T = 300$ K, $P = 101,325$ Pa, and $R = 8.314$ J/(mol · K), the gas volume is closest to:
- (A) 24.6 m³ (B) 22.4 m³
(C) 0.0246 m³ (D) 0.0492 m³
109. A Carnot engine operates between a cold reservoir at 300 K and a hot reservoir at 600 K. Its maximum (Carnot) efficiency is:
- (A) 0.50 (B) 0.25
(C) 0.75 (D) 1.0
110. A power cycle produces net work of 400 kJ from a heat input of 1000 kJ. Its thermal efficiency is:
- (A) 0.60 (B) 0.25
(C) 0.40 (D) 1.0

Answer Key

1	2	3	4	5	6	7	8	9	10
B	A	C	D	A	B	B	D	A	B

11	12	13	14	15	16	17	18	19	20
C	A	B	C	A	B	D	A	B	C

21	22	23	24	25	26	27	28	29	30
D	A	B	C	A	D	C	A	B	D

31	32	33	34	35	36	37	38	39	40
A	C	B	A	C	D	B	A	C	D

41	42	43	44	45	46	47	48	49	50
A	C	B	A	D	C	B	A	D	C

51	52	53	54	55	56	57	58	59	60
A	B	C	A	B	C	A	D	B	C

61	62	63	64	65	66	67	68	69	70
A	D	B	C	A	D	A	B	C	D

71	72	73	74	75	76	77	78	79	80
A	C	B	D	A	C	B	D	B	A

81	82	83	84	85	86	87	88	89	90
C	D	A	B	A	B	C	A	D	B

91	92	93	94	95	96	97	98	99	100
C	A	B	D	A	C	B	A	D	C

101	102	103	104	105	106	107	108	109	110
B	A	C	D	B	C	A	D	A	C

Worked Solutions

1. **Answer: B.** $p = \rho gh = 1000(9.81)(6) = 58,860 \text{ Pa} \approx 58.9 \text{ kPa}$. Handbook: Fluid Mechanics — hydrostatic pressure
2. **Answer: A.** $A = \frac{\pi}{4}(0.40)^2 = 0.1257 \text{ m}^2$; $V = Q/A = 0.30/0.1257 = 2.39 \text{ m/s}$. Handbook: Fluid Mechanics — continuity
3. **Answer: C.** $V = \sqrt{2gh} = \sqrt{2(9.81)(5)} = \sqrt{98.1} = 9.90 \text{ m/s}$. Handbook: Fluid Mechanics — Bernoulli/Torricelli
4. **Answer: D.** $Re = VD/\nu = (2.0)(0.10)/(1.0 \times 10^{-6}) = 2.0 \times 10^5$. Handbook: Fluid Mechanics — Reynolds number
5. **Answer: A.** $h_f = f \frac{L}{D} \frac{V^2}{2g} = 0.02 \frac{100}{0.20} \frac{3^2}{2(9.81)} = 10(0.459) = 4.59 \text{ m}$. Handbook: Fluid Mechanics — Darcy-Weisbach
6. **Answer: B.** $V = \frac{1}{n} R_h^{2/3} S^{1/2} = \frac{1}{0.013} (0.50)^{2/3} (0.001)^{1/2} = 76.9(0.630)(0.0316) = 1.53 \text{ m/s}$. Handbook: Fluid Mechanics — Manning's equation
7. **Answer: B.** $V = \frac{1}{0.015} (0.50)^{2/3} (0.0009)^{1/2} = 66.7(0.630)(0.030) = 1.26 \text{ m/s}$; $A = 2.0(1.0) = 2.0 \text{ m}^2$; $Q = VA = 1.26(2.0) = 2.52 \text{ m}^3/\text{s}$. Handbook: Hydraulics — Manning open channel
8. **Answer: D.** $P = \frac{\rho g Q H}{\eta} = \frac{1000(9.81)(0.05)(20)}{0.75} = \frac{9810}{0.75} = 13,080 \text{ W} \approx 13.1 \text{ kW}$. Handbook: Fluid Mechanics — pump power
9. **Answer: A.** Pumps in series add heads at equal flow: $H = 15 + 15 = 30 \text{ m}$. Handbook: Hydraulics — pumps in series
10. **Answer: B.** $Q = 1.84 L H^{3/2} = 1.84(2.0)(0.30)^{1.5} = 3.68(0.1643) = 0.605 \text{ m}^3/\text{s}$. Handbook: Hydraulics — rectangular weir
11. **Answer: C.** $Q = C_d A \sqrt{2gh} = 0.61(0.01)\sqrt{2(9.81)(3)} = 0.0061(7.67) = 0.0468 \text{ m}^3/\text{s}$. Handbook: Hydraulics — orifice discharge
12. **Answer: A.** $h_L = K \frac{V^2}{2g} = 1.5 \frac{4^2}{2(9.81)} = 1.5(0.815) = 1.22 \text{ m}$. Handbook: Fluid Mechanics — minor losses
13. **Answer: B.** $F = pA = 50,000(0.20) = 10,000 \text{ N} = 10.0 \text{ kN}$. Handbook: Fluid Mechanics — hydrostatic force
14. **Answer: C.** $\eta = \frac{C_{in} - C_{out}}{C_{in}} = \frac{200 - 20}{200} = 0.90 = 90\%$. Handbook: Water/Wastewater — removal efficiency
15. **Answer: A.** $t = V/Q = 2000/500 = 4 \text{ h}$. Handbook: Water/Wastewater — detention time
16. **Answer: B.** $SOR = Q/A = 4000/200 = 20 \text{ m/d}$. Handbook: Water/Wastewater — overflow rate

17. **Answer: D.** Ca: $40 \times (50/20) = 100$; Mg: $12 \times (50/12.15) = 49.4$; total ≈ 149 mg/L as CaCO_3 . Handbook: Water — hardness as CaCO_3
18. **Answer: A.** $CT = C \times t = 2.0(30) = 60$ mg · min/L. Handbook: Water — disinfection CT
19. **Answer: B.** BOD mass load = $QS_0 = 10,000(200)/1000 = 2000$ kg/d; MLSS mass = $VX = 2500(2000)/1000 = 5000$ kg; $F/M = 2000/5000 = 0.40$ d⁻¹. Handbook: Wastewater — activated sludge F/M
20. **Answer: C.** $\text{SRT} = \frac{\text{solids in reactor}}{\text{solids wasted per day}} = \frac{5000}{500} = 10$ d. Handbook: Wastewater — solids retention time
21. **Answer: D.** $\text{BOD}_t = L_0(1 - e^{-kt}) = 300(1 - e^{-0.23(5)}) = 300(1 - 0.317) = 205$ mg/L. Handbook: Wastewater — BOD decay
22. **Answer: A.** $C = \frac{Q_1C_1 + Q_2C_2}{Q_1 + Q_2} = \frac{2(10) + 3(20)}{5} = \frac{80}{5} = 16$ mg/L. Handbook: Water — mass balance blending
23. **Answer: B.** lb/d = (mg/L)(MGD)(8.34) = $2.0(5.0)(8.34) = 83.4$ lb/d. Handbook: Water — chlorine dose (lb/d)
24. **Answer: C.** meq/L = (mg/L as CaCO_3)/50 = $150/50 = 3$ meq/L. Handbook: Water — hardness equivalents
25. **Answer: A.** Wet mass = $1000/0.05 = 20,000$ kg; volume = $20,000/1000 = 20$ m³/d. Handbook: Wastewater — sludge volume
26. **Answer: D.** lb/d = $(250)(0.50)(8.34) = 1043$ lb/d. Handbook: Wastewater — BOD mass load
27. **Answer: C.** $Q = CiA = 0.50(2.0)(10) = 10$ cfs (1 acre-in/hr $\approx$ 1 cfs). Handbook: Hydrology — rational method
28. **Answer: A.** $Q = CiA/360 = 0.40(90)(5)/360 = 180/360 = 0.50$ m³/s. Handbook: Hydrology — rational method (SI)
29. **Answer: B.** $t = L/V = 300/1.5 = 200$ s = 3.33 min. Handbook: Hydrology — time of concentration
30. **Answer: D.** $\Delta S = (Q_{in} - Q_{out})t = (5 - 2)(3600) = 3(3600) = 10,800$ m³. Handbook: Hydrology — detention storage
31. **Answer: A.** $D_0 = DO_{sat} - DO = 9.0 - 6.0 = 3.0$ mg/L. Handbook: Hydrology — DO deficit
32. **Answer: C.** $D = \frac{k_d L_0}{k_r - k_d} (e^{-k_d t} - e^{-k_r t}) + D_0 e^{-k_r t} = \frac{0.30(20)}{0.10} (e^{-0.6} - e^{-0.8}) + 2e^{-0.8} = 60(0.0995) + 0.899 = 6.87$ mg/L. Handbook: Hydrology — Streeter-Phelps
33. **Answer: B.** $t_c = \frac{1}{k_r - k_d} \ln \left[\frac{k_r}{k_d} \left(1 - \frac{D_0(k_r - k_d)}{k_d L_0} \right) \right] = \frac{1}{0.10} \ln[1.333(0.967)] = 10 \ln(1.289) = 2.54$ d. Handbook: Hydrology — critical deficit time

34. **Answer: A.** $P = ET + R + \Delta S \Rightarrow \Delta S = 1000 - 600 - 300 = 100$ mm. Handbook: Hydrology — water budget
35. **Answer: C.** $\text{depth} = i \times t = 50(0.50) = 25$ mm. Handbook: Hydrology — IDF/rainfall depth
36. **Answer: D.** $V = CPA = 0.60(0.050)(10,000) = 300$ m³. Handbook: Hydrology — runoff volume
37. **Answer: B.** $q = Ki = (1.0 \times 10^{-4})(0.02) = 2.0 \times 10^{-6}$ m/s. Handbook: Groundwater — Darcy's law
38. **Answer: A.** $v_s = q/n = (2.0 \times 10^{-6})/0.25 = 8.0 \times 10^{-6}$ m/s. Handbook: Groundwater — seepage velocity
39. **Answer: C.** $i = \Delta h/L = 2.0/500 = 0.004$. Handbook: Groundwater — hydraulic gradient
40. **Answer: D.** $T = Kb = 15(20) = 300$ m²/d. Handbook: Groundwater — transmissivity
41. **Answer: A.** $Q = KiA = (1.0 \times 10^{-3})(0.01)(50) = 5.0 \times 10^{-4}$ m³/s. Handbook: Groundwater — Darcy flow rate
42. **Answer: C.** $Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)} = \frac{2\pi(300)(3)}{\ln 10} = \frac{5655}{2.303} = 2456$ m³/d. Handbook: Groundwater — Thiem equation
43. **Answer: B.** $R = 1 + \frac{\rho_b K_d}{n} = 1 + \frac{1.6(2.0)}{0.30} = 1 + 10.67 = 11.67$. Handbook: Groundwater — retardation factor
44. **Answer: A.** flow per unit width = $qb = (2.0 \times 10^{-6})(20) = 4.0 \times 10^{-5}$ m²/s. Handbook: Groundwater — unit-width discharge
45. **Answer: D.** $t = L/v_s = 80/(8.0 \times 10^{-6}) = 1.0 \times 10^7$ s = 115.7 d. Handbook: Groundwater — contaminant travel time
46. **Answer: C.** $\mu\text{g}/\text{m}^3 = \frac{\text{ppm} \cdot MW \cdot 1000}{24.45} = \frac{0.10(64)(1000)}{24.45} = 262$ $\mu\text{g}/\text{m}^3$. Handbook: Air — ppm to mass concentration
47. **Answer: B.** $\text{ppm} = \frac{(\mu\text{g}/\text{m}^3)(24.45)}{MW \cdot 1000} = \frac{5725(24.45)}{28(1000)} = 5.0$ ppm. Handbook: Air — mass concentration to ppm
48. **Answer: A.** rate = EF $\times$ activity = 10(500) = 5000 kg/d. Handbook: Air — emission factor
49. **Answer: D.** $\Delta T = (9.8)(2.0) = 19.6$ °C cooling. Handbook: Air — dry adiabatic lapse rate
50. **Answer: C.** $\eta = 1 - e^{-wA/Q} = 1 - e^{-(0.10)(200)/10} = 1 - e^{-2} = 1 - 0.135 = 0.865$. Handbook: Air — ESP Deutsch equation
51. **Answer: A.** $A = -\frac{Q}{w} \ln(1 - \eta) = -\frac{10}{0.10} \ln(0.01) = -100(-4.605) = 461$ m². Handbook: Air — ESP design area

52. **Answer: B.** $A/C \text{ ratio} = Q/A = 600/300 = 2.0 \text{ m/min.}$ Handbook: Air — baghouse
air-to-cloth ratio
53. **Answer: C.** $C = \frac{2.0 \text{ g/s}}{100 \text{ m}^3/\text{s}} = 0.020 \text{ g/m}^3 = 20 \text{ mg/m}^3.$ Handbook: Air — plume dilution
54. **Answer: A.** $C = \frac{Q}{\pi \sigma_y \sigma_z u} = \frac{100}{\pi(30)(15)(4)} = \frac{100}{5655} = 0.0177 \text{ g/m}^3.$ Handbook: Air —
Gaussian dispersion
55. **Answer: B.** $M = \frac{100 \text{ mg/L}}{100 \text{ g/mol} \times 1000 \text{ mg/g}} = 1.0 \times 10^{-3} \text{ mol/L.}$ Handbook: Chemistry —
mg/L to molarity
56. **Answer: C.** $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(10^{-4}) = 4.$ Handbook: Chemistry — pH
57. **Answer: A.** $\text{pOH} = 14 - \text{pH} = 14 - 9 = 5.$ Handbook: Chemistry — pOH
58. **Answer: D.** $[\text{OH}^-] = 10^{-\text{pOH}} = 10^{-5} \text{ mol/L.}$ Handbook: Chemistry — hydroxide concentration
59. **Answer: B.** $t_{1/2} = 0.693/k = 0.693/0.05 = 13.86 \text{ d.}$ Handbook: Chemistry — first-order
half-life
60. **Answer: C.** $C = C_0 e^{-kt} = 50e^{-0.10(10)} = 50e^{-1} = 50(0.368) = 18.4 \text{ mg/L.}$ Handbook:
Chemistry — first-order kinetics
61. **Answer: A.** $C_g = H_c C_l = 0.10(2.0) = 0.2 \text{ (gas-phase value, same units basis).}$ Handbook:
Chemistry — Henry's law
62. **Answer: D.** 1 mol CH_4 needs 2 mol $\text{O}_2 = 2(32) = 64 \text{ g O}_2.$ Handbook: Chemistry —
combustion stoichiometry
63. **Answer: B.** $P = P_0 + kt = 10,000 + 200(10) = 12,000.$ Handbook: Principles — arithmetic
population growth
64. **Answer: C.** $P = P_0(1 + r)^t = 10,000(1.03)^{10} = 10,000(1.3439) = 13,439.$ Handbook:
Principles — geometric growth
65. **Answer: A.** $\frac{4,000,000 \text{ gpd}}{20,000 \text{ people}} = 200 \text{ gpcd.}$ Handbook: Principles — per-capita demand
66. **Answer: D.** PFR: $C/C_0 = e^{-k\tau} = e^{-0.5(2)} = e^{-1} = 0.368.$ Handbook: Principles — PFR
first-order
67. **Answer: A.** CSTR: $C/C_0 = \frac{1}{1 + k\tau} = \frac{1}{1 + 0.5(2)} = \frac{1}{2} = 0.50.$ Handbook: Principles — CSTR
first-order
68. **Answer: B.** $\sigma = F/A = 10,000/0.0020 = 5.0 \times 10^6 \text{ Pa} = 5.0 \text{ MPa.}$ Handbook: Principles —
axial stress
69. **Answer: C.** $\varepsilon = \sigma/E = (200 \times 10^6)/(200 \times 10^9) = 1.0 \times 10^{-3}.$ Handbook: Principles —
Hooke's law/strain

70. **Answer: D.** $Q = 200 \text{ gpcd} \times 5000 = 1,000,000 \text{ gpd} = 1.0 \text{ MGD}$. Handbook: Principles — design flow
71. **Answer: A.** $5.0 \text{ lb/cap/d} \times 10,000 = 50,000 \text{ lb/d} = 25 \text{ tons/d}$. Handbook: Solid Waste — MSW generation
72. **Answer: C.** $V_c/V_t = \rho_t/\rho_c = 100/500 = 0.20$ (80% reduction). Handbook: Solid Waste — compaction ratio
73. **Answer: B.** $\text{mass} = 1000(100) = 100,000 \text{ kg}$; $V_c = 100,000/500 = 200 \text{ m}^3$. Handbook: Solid Waste — compacted volume
74. **Answer: D.** $\text{CH}_4 = 0.50(200) = 100 \text{ m}^3$. Handbook: Solid Waste — landfill gas composition
75. **Answer: A.** infiltration depth $= 1.0 - 0.40 - 0.20 = 0.40 \text{ m}$; $V = 0.40(1000) = 400 \text{ m}^3$. Handbook: Solid Waste — leachate water balance
76. **Answer: C.** $8/40 = 0.20 = 20\%$. Handbook: Solid Waste — diversion/recycling rate
77. **Answer: B.** $10 \text{ ton} \times 2000 \text{ lb/ton} \times 5000 \text{ BTU/lb} = 1.0 \times 10^8 \text{ BTU}$. Handbook: Solid Waste — energy content
78. **Answer: D.** airspace $= 100/0.50 = 200 \text{ m}^3/\text{d}$. Handbook: Solid Waste — landfill airspace
79. **Answer: B.** $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$; roots 2 and 3; larger is 3. Handbook: Mathematics — quadratic roots
80. **Answer: A.** $\sin 30^\circ + \cos 60^\circ = 0.5 + 0.5 = 1.0$. Handbook: Mathematics — trigonometry
81. **Answer: C.** $f'(x) = 6x$; at $x = 2$, $f' = 12$. Handbook: Mathematics — derivative
82. **Answer: D.** $\int_0^2 3x^2 dx = [x^3]_0^2 = 8$. Handbook: Mathematics — definite integral
83. **Answer: A.** $d = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. Handbook: Mathematics — distance formula
84. **Answer: B.** $2^x = 16 = 2^4 \Rightarrow x = 4$. Handbook: Mathematics — exponential equation
85. **Answer: A.** The NCEES Model Rules require engineers to hold paramount the safety, health, and welfare of the public. Handbook: Ethics — public welfare paramount
86. **Answer: B.** Engineers must perform services only in areas of their competence. Handbook: Ethics — competence
87. **Answer: C.** The Clean Water Act regulates pollutant discharges to navigable surface waters (NPDES). Handbook: Ethics/Regulations — CWA
88. **Answer: A.** CERCLA (Superfund) addresses cleanup of and liability for releases at abandoned hazardous-waste sites; RCRA covers active cradle-to-grave management. Handbook: Regulations — CERCLA vs RCRA
89. **Answer: D.** The engineer must notify the proper authorities and the client as required to protect the public when public safety is endangered. Handbook: Ethics — duty to protect public

90. **Answer: B.** NEPA requires an Environmental Impact Statement for major federal actions significantly affecting the environment. Handbook: Regulations — NEPA
91. **Answer: C.** $P = F(1 + i)^{-n} = 10,000(1.05)^{-10} = 10,000(0.6139) = \6139 . Handbook: Economics — P/F factor
92. **Answer: A.** $P = A \frac{(1 + i)^n - 1}{i(1 + i)^n} = 1000 \frac{(1.08)^5 - 1}{0.08(1.08)^5} = 1000(3.993) = \3993 . Handbook: Economics — P/A factor
93. **Answer: B.** $A = P \frac{i(1 + i)^n}{(1 + i)^n - 1} = 20,000 \frac{0.06(1.06)^{10}}{(1.06)^{10} - 1} = 20,000(0.1359) = \2717 . Handbook: Economics — A/P capital recovery
94. **Answer: D.** $d = \frac{C - S}{N} = \frac{50,000 - 5000}{9} = \frac{45,000}{9} = \$5000/\text{yr}$. Handbook: Economics — straight-line depreciation
95. **Answer: A.** $B/C = 150,000/100,000 = 1.5$. Handbook: Economics — benefit-cost ratio
96. **Answer: C.** $Q^* = \frac{FC}{p - v} = \frac{10,000}{5 - 3} = 5000$ units. Handbook: Economics — break-even
97. **Answer: B.** $\bar{x} = \frac{2 + 4 + 4 + 4 + 5 + 5 + 7 + 9}{8} = \frac{40}{8} = 5$. Handbook: Statistics — sample mean
98. **Answer: A.** $\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{N}} = \sqrt{\frac{32}{8}} = \sqrt{4} = 2.0$. Handbook: Statistics — population standard deviation
99. **Answer: D.** $z = \frac{x - \mu}{\sigma} = \frac{70 - 50}{10} = 2.0$. Handbook: Statistics — z-score
100. **Answer: C.** $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{32}{7}} = \sqrt{4.571} = 2.14$. Handbook: Statistics — sample standard deviation
101. **Answer: B.** $\text{CDI} = \frac{C \cdot IR}{BW} = \frac{0.050(2.0)}{70} = 0.00143 \text{ mg/kg-d}$. Handbook: Risk — chronic daily intake
102. **Answer: A.** $\text{Risk} = \text{CDI} \times SF = 0.00143(0.50) = 7.1 \times 10^{-4}$. Handbook: Risk — cancer risk
103. **Answer: C.** $HQ = \frac{\text{intake}}{RfD} = \frac{0.0020}{0.010} = 0.20$. Handbook: Risk — hazard quotient
104. **Answer: D.** $\text{intake} = C \times IR = 10(2.0) = 20 \text{ mg/d}$. Handbook: Risk — daily intake
105. **Answer: B.** $\text{CF} = \text{activity} \times \text{EF} = 1000(0.50) = 500 \text{ kg CO}_2$. Handbook: Energy — carbon footprint
106. **Answer: C.** $\text{CO}_2\text{e} = 2.0 \times 25 = 50 \text{ kg CO}_2\text{e}$. Handbook: Energy — GWP/CO₂-equivalent
107. **Answer: A.** $E = 100 \text{ L} \times 36 \text{ MJ/L} = 3600 \text{ MJ}$. Handbook: Energy — fuel energy content

108. **Answer: D.** $V = \frac{nRT}{P} = \frac{2.0(8.314)(300)}{101,325} = \frac{4988}{101,325} = 0.0492 \text{ m}^3$. Handbook:
Thermodynamics — ideal gas law
109. **Answer: A.** $\eta_{Carnot} = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{600} = 0.50$. Handbook: Thermodynamics — Carnot
efficiency
110. **Answer: C.** $\eta_{th} = \frac{W_{net}}{Q_{in}} = \frac{400}{1000} = 0.40$. Handbook: Thermodynamics — thermal efficiency